Components and Props

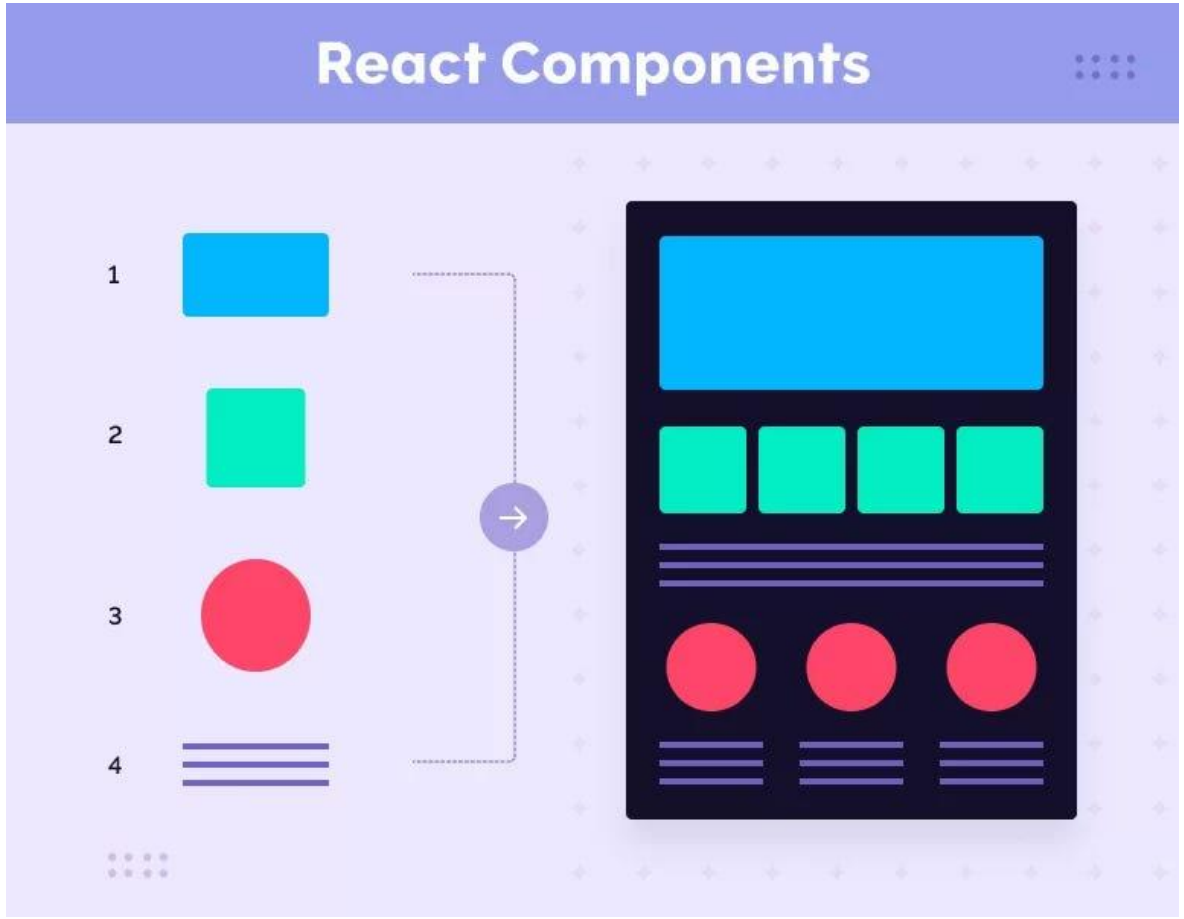
Overview

- React components
- Functional components
- Props

React Components

- Using React, we aim to split the user interface into reusable **components**

- DRY principle (Don’t Repeat Yourself)



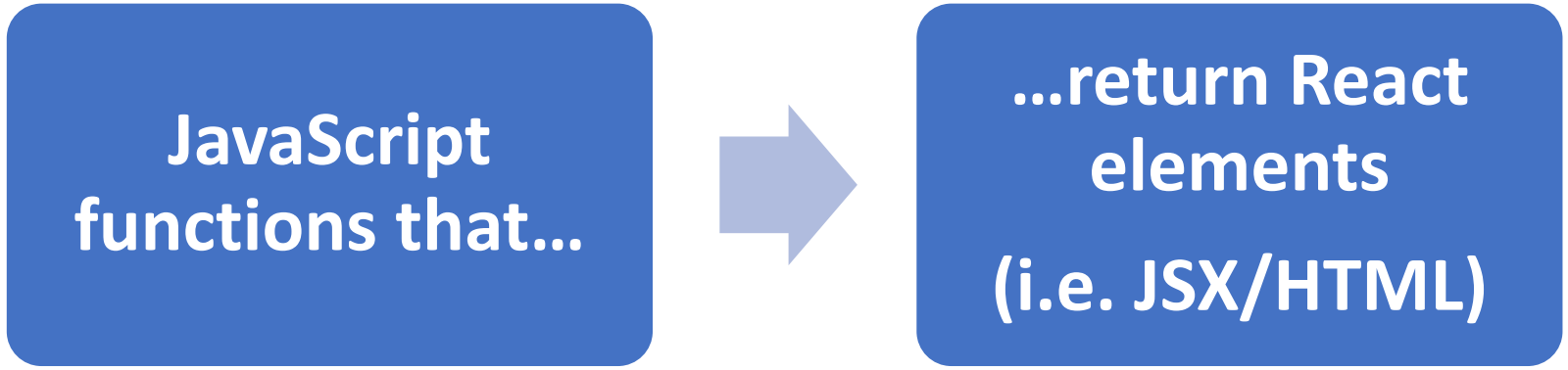
React Components

- “Conceptually, components are like **JavaScript functions**. They accept arbitrary inputs (called ‘props’) and **return React elements** describing what should appear on the screen.” (reactjs.org)

- “Components are **independent** and **reusable** bits of code. They serve the same purpose as **JavaScript functions**, but work in isolation and **return HTML.**” (W3Schools.com)

React Components

- So React components are:



- A simple component will look just like a JavaScript function with some HTML inside

React component

```
1  import React from "react";
2
3  function FunctionalComponent() {
4   return <h1>Hello, world</h1>;
5  }
```

- Note the line `import React from “react”` – this is necessary in order to use JSX within the component

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Functional components

- A functional component looks just like a JavaScript function

- Note the line `import React from “react”` – this is necessary in order to use JSX within the component

```
1  import React from "react";
2
3  function FunctionalComponent() {
4   return <h1>Hello, world</h1>;
5  }
```

Creating a functional component

```
import React from 'react';

const Subject = () => {

    return (
        <p>
          This is a subject!
        </p>
    )
}

export default Subject;
```

Import React library from React package

Create subject component (using arrow function syntax)

Return a paragraph (using JSX)

Export the subject component

Using the component

```
import React from 'react';
import './App.css';
import Subject from './components/subject';

function App() {

  return (
    <div className="App">
      <h1>Hi, I'm a React App</h1>
      <Subject />
    </div>
  );
}

export default App;
```

Import Subject component from its file
Note the component name: always use an uppercase first letter

Use subject component (with no contents):
<Subject />

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Props

- **Props** are properties that we can pass into our components

```
return (
  <div className="App">
    <h1>Hi, I'm a React App</h1>
    <Subject title = "Web App Dev 2" year = "Year 3" />
    <Subject title = "Digital Graphic Design" year = "Year 3" />
    <Subject  title = "NoSQL Databases" year = "Year 3" />
  </div>
);
```

Using props

- When our component takes in a props parameter, we can use the properties with the syntax **{props.propertyname}**

```
const subject = (props) => {

    return (
        <div>
            <p>
                This is a subject! Its title is {props.title} and it is in {props.
                year} of the course.
            </p>
        </div>
    )
}
```

The children prop

- The children prop is a special property that holds the contents of any of our component elements

```
return (
  <div className="App">
    <h1>Hi, I'm a React App</h1>
    <Subject title = "Web App Dev 2" year = "Year 3">
      Read more about Web App Dev 2
    </Subject>
    <Subject title = "Digital Graphic Design" year = "Year 3" />
    <Subject  title = "NoSQL Databases" year = "Year 3" />
  </div>
);
```

Using the children prop

- In our component, we can access the children prop using the syntax **{props.children}**

```
return (
    <div>
        <p>
            This is a subject! Its title is {props.title} and it is in {props.
            year} of the course.
        </p>
        <p>
            {props.children}
        </p>
    </div>
)
```

Quiz!